

Workbook: The Heart

Week 3 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on the gross anatomy of the heart: its position, the pericardium and heart wall layers, the four chambers, the four valves, the great vessels, the coronary circulation, and clinical anatomy of valve problems and myocardial infarction.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Position and pericardium · 4 questions

Q01 · DOK 1 · Recall

Describe the position of the heart in the thorax.

Q02 · DOK 1 · Recall

Name the layers of the pericardium from outside in.

Q03 · DOK 2 · Apply

Why does the heart need pericardial fluid, and what happens if too much accumulates?

Q04 • DOK 1 • Recall

Name the three layers of the heart wall from outside in and describe each.

Chambers • 5 questions**Q05 • DOK 1 • Recall**

Name the four chambers of the heart and their general role.

Q06 • DOK 2 • Compare

Compare atrial and ventricular walls. Why is the difference in thickness functional?

Q07 • DOK 1 • Recall

What is the interventricular septum, and why does it matter functionally?

Q08 • DOK 1 • Recall

What is the foramen ovale before birth, and what does it become after birth?

Q09 · DOK 1 · Recall

What are papillary muscles and chordae tendineae, and what do they do together?

Valves · 4 questions**Q10 · DOK 1 · Recall**

Name the four heart valves, the chambers or vessels they sit between, and the basic function of all heart valves.

Q11 · DOK 2 · Compare

Compare the structure of atrioventricular and semilunar valves.

Q12 · DOK 1 · Recall

What is the cardiac skeleton, and what does it do for the valves?

Q13 · DOK 2 · Apply

What produces the lub-dub of the heart sounds?

Great vessels · 4 questions

Q14 • DOK 2 • Apply

Name the great vessels that enter or leave each chamber, with the right side first.

Q15 • DOK 1 • Recall

Trace blood through the pulmonary circuit.

Q16 • DOK 1 • Recall

Trace blood through the systemic circuit.

Q17 • DOK 2 • Apply

The pulmonary arteries carry deoxygenated blood and pulmonary veins carry oxygenated blood. Why does that make sense even though it sounds backward?

Coronary circulation • 4 questions**Q18 • DOK 1 • Recall**

Name the two main coronary arteries and the major branches of each.

Q19 • DOK 2 • Apply

Which parts of the heart does each main coronary artery typically supply?

Q20 • DOK 1 • Recall

Where does coronary venous blood drain?

Q21 • DOK 3 • Reason

A patient has chest pain. ECG shows changes in leads that correspond to the anterior wall of the left ventricle. Predict which coronary artery is most likely blocked and why.

Clinical anatomy • 2 questions**Q22 • DOK 3 • Reason**

A myocardial infarction kills cardiac muscle in part of the ventricular wall. Using what you know about cardiac muscle's regenerative capacity, predict the structural outcome.

Q23 · DOK 3 · Reason

In mitral valve prolapse, the cusps bulge back into the left atrium during ventricular contraction. Predict what blood does and what symptom the patient might notice.
